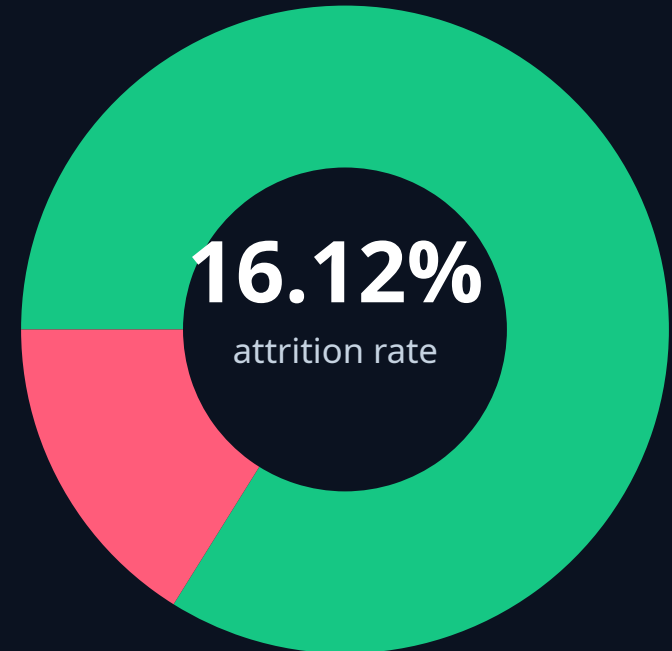


Predicting Employee Attrition Risk

IBM HR Analytics Employee Attrition & Performance dataset

The DataSmiths



1,470

employee records

35

original columns

0

missing values / duplicates

237

employees left

Executive summary

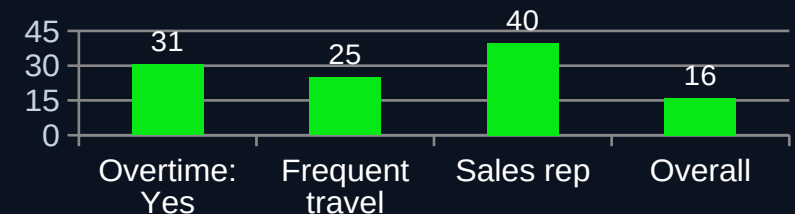
The data points to specific, addressable retention pressures.

Employees do not leave evenly across the workforce.

- 01 Workload pressure** Overtime employees show almost 3× the attrition rate of non-overtime employees.
- 02 Mobility pressure** Frequent business travel is associated with the highest travel-related attrition.
- 03 Role concentration** Sales Representative and Laboratory Technician roles carry the clearest role-level risk.
- 04 Model limitation** The selected model identified about one-third of employees who left, so HR review remains essential.

Main action

Use the model as an early-warning system, then combine it with targeted workload, travel and career-growth interventions.



Business problem & objective

A practical HR analytics problem: predict who may leave and explain why.

Attrition is not just a headcount issue. It affects continuity, recruitment cost, productivity and team knowledge.

Question 1

Which employees are more likely to leave?

1

EDA

Find patterns

Question 2

Which factors are most associated with turnover?

2

Model

Predict risk

3

Interpret

Explain drivers

Question 3

What retention actions should the organization prioritize?

4

Recommended

Reduce attrition

Dataset & workflow

The team prepared a clean, model-ready dataset before building the classifier.

1,470

rows / employee records

35

original columns

0

missing values

0

duplicate rows

Removed non-informative fields: EmployeeCount, Over18, StandardHours and EmployeeNumber.

Clean & validate

standardize text
check missing/duplicates

Explore

attrition patterns
by role, workload &
demographics

Encode

binary target
one-hot categorical
features

Train & test

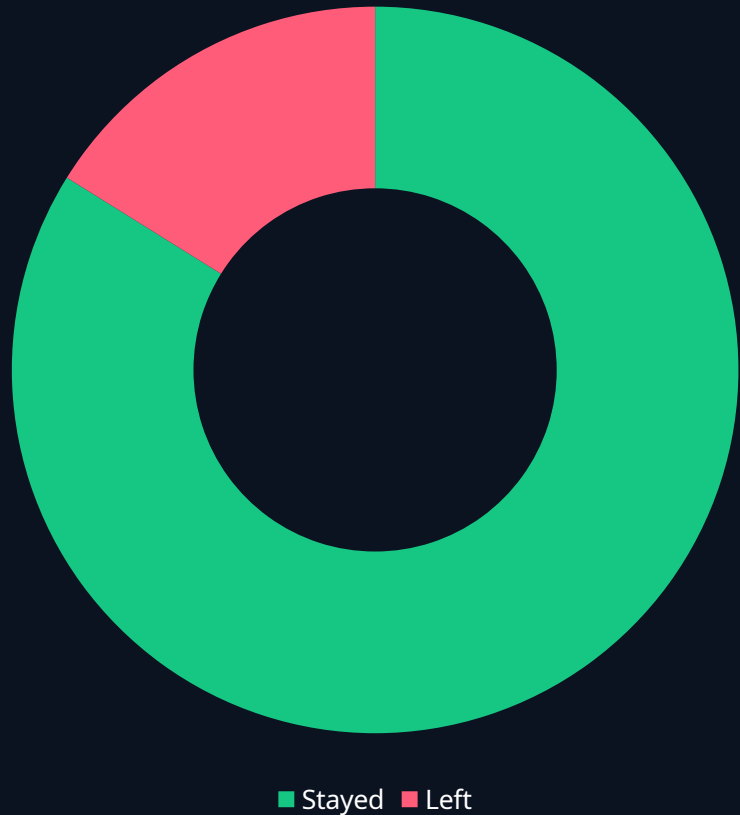
stratified split
classification models

Interpret

feature drivers
retention
recommendations

Attrition baseline

The target is imbalanced, so evaluation must look beyond accuracy.



83.88% stayed

1,233 employees

16.12% left

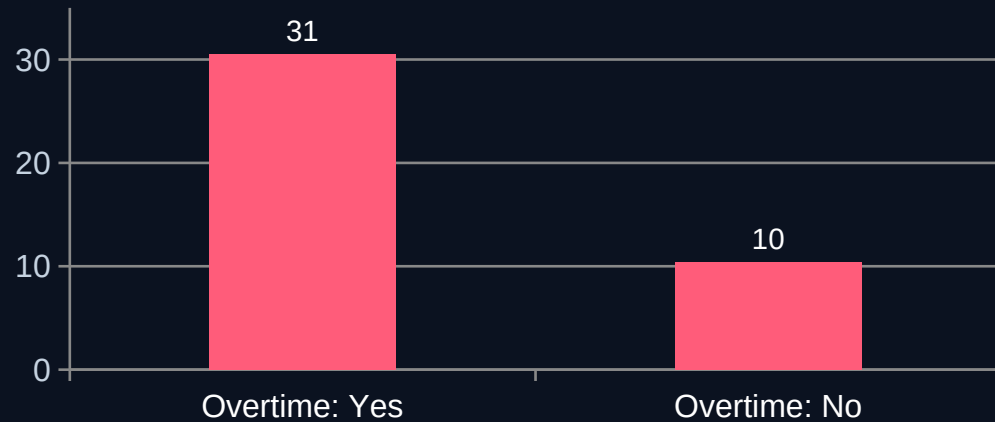
237 employees

Modeling implication

A model that predicts “no attrition” for everyone would appear accurate, but would fail the business objective. Recall and F1-score matter.

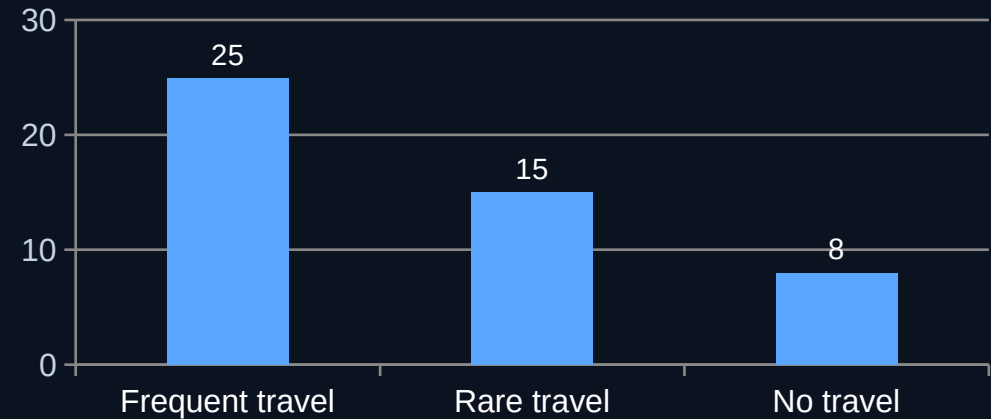
EDA: workload and travel signals

Attrition rises sharply where work pressure and mobility demands increase.



Workload signal

Employees working overtime have a 30.5% attrition rate versus 10.4% for those not working overtime.

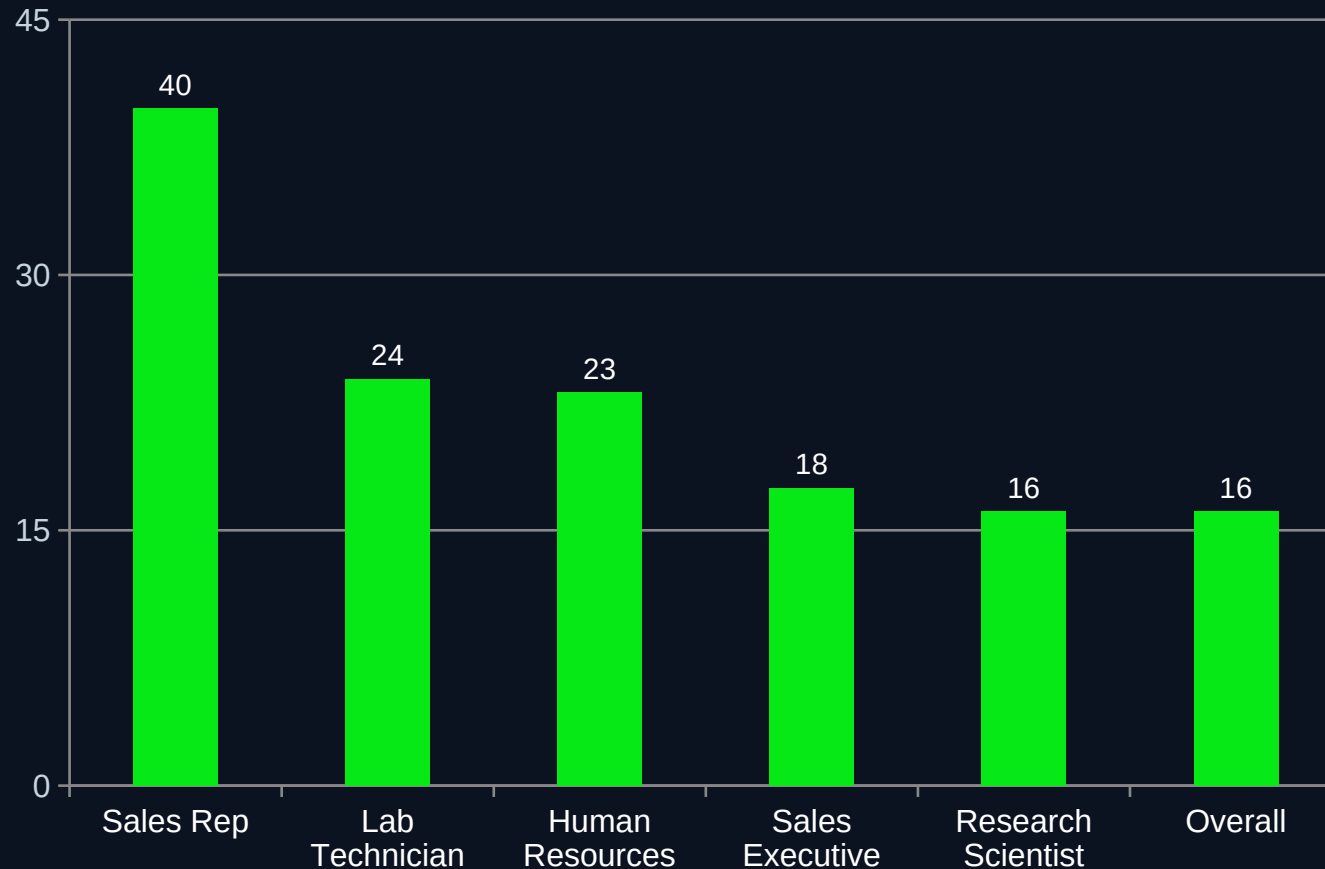


Travel signal

Frequent travelers have the highest travel-related attrition rate at 24.9%.

EDA: role-level risk concentration

Attrition is highest in specific job roles, not evenly distributed across the organization.



Highest risk role

Sales Representatives have the highest role-level attrition rate at 39.8%.

Operational meaning

Retention should be targeted. A single company-wide intervention may miss the teams where attrition is most concentrated.

Modeling approach & evaluation

The selected model balances predictive performance with interpretability.

Preprocessing pipeline

- Drop** ID + constant fields
- Encode** categorical variables
- Scale** for Logistic Regression
- Train** stratified split

Model	Accuracy	Recall	F1	ROC-AUC
Logistic Regression	0.861	0.340	0.438	0.808
Random Forest	0.833	0.064	0.109	0.786
Decision Tree	0.759	0.362	0.321	0.598

Confusion matrix — Logistic Regression

	Pred: Stay	Pred: Leave
Actual Stay	237	10
Actual Leave	31	16

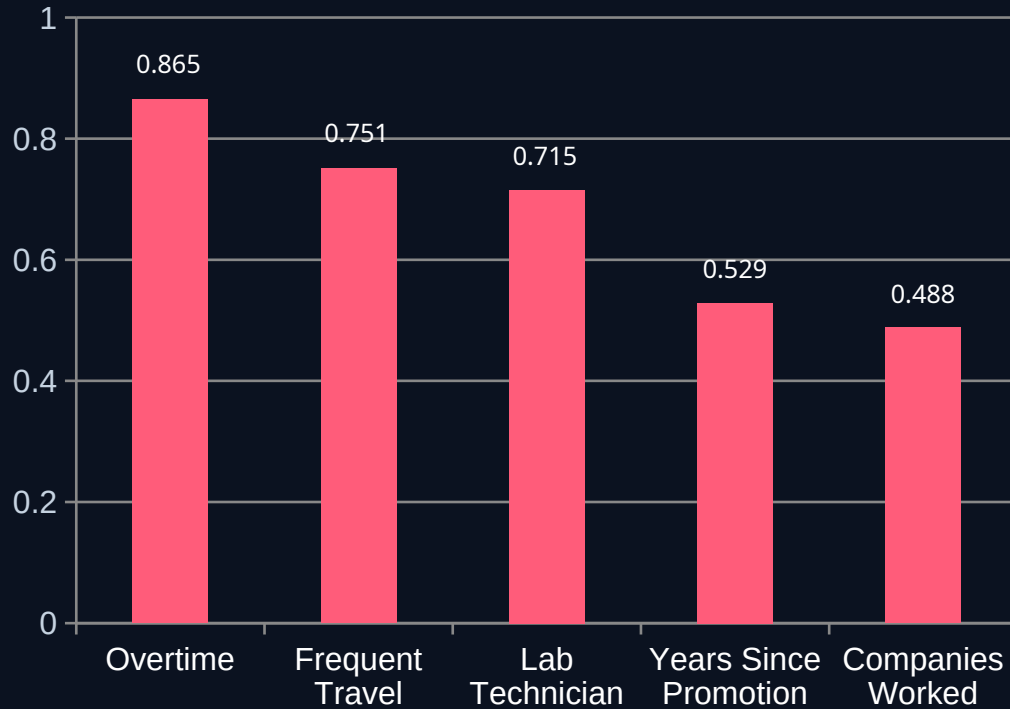
Why Logistic Regression?

It achieved the highest accuracy, precision, F1 and ROC-AUC. Decision Tree had slightly higher recall, but weaker overall performance.

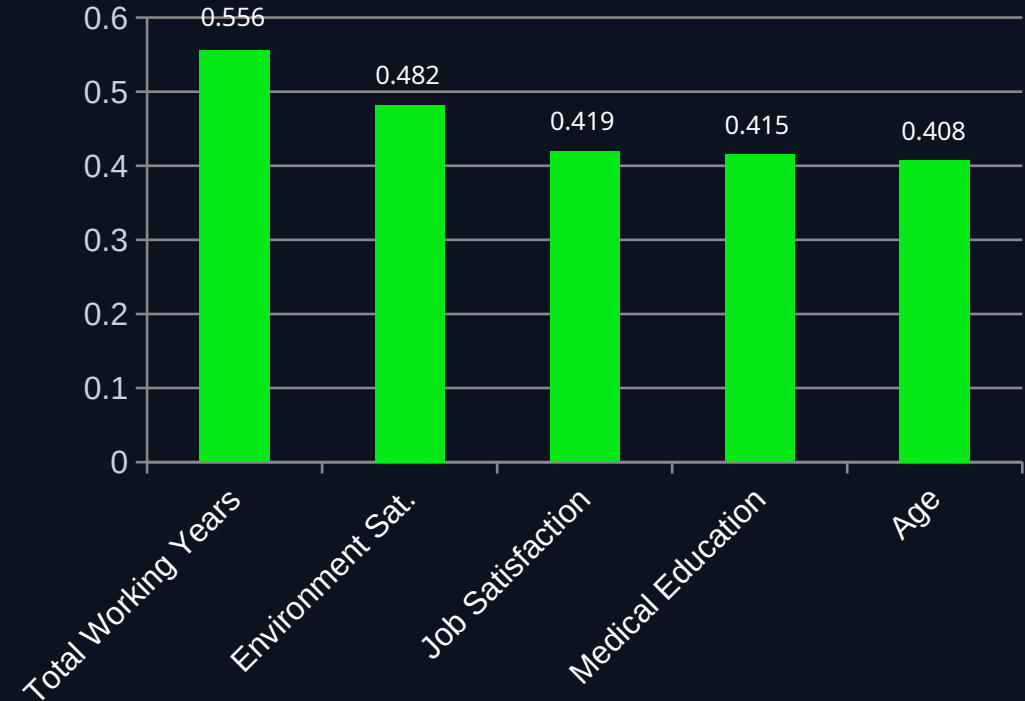
Interpreting the model

The strongest drivers reinforce what the EDA suggested.

Top risk-increasing coefficients



Top protective coefficients



Interpretation: workload, travel and role context raise predicted risk; experience and satisfaction lower it. These are associations, not proof of cause.

Recommendations & next steps

Turn the analysis into a practical retention program.

1 Manage overtime risk

Audit overtime-heavy teams; rotate workload; flag sustained overtime before it becomes resignation risk.

3 Prioritize high-risk roles

Start with Sales Representatives, Laboratory Technicians and HR roles where attrition is concentrated.

2 Support frequent travelers

Review travel expectations, recovery time, allowances and manager check-ins for frequent travelers.

4 Strengthen career growth & support

Use clear promotion paths, regular manager check-ins and satisfaction feedback, especially for younger or lower-income employees.

MODEL USE

Use predictions as an early-warning signal alongside employee feedback and human judgement—not as the sole basis for employment decisions.